My Equation — Statistics Screening Assignment (Pre-Joining Interns)

Purpose

Assess your current statistics and data-thinking skills using realistic My Equation scenarios (Robo Ai, Robopack, E-Drive, B2B college tie-ups). Keep answers concise, show workings, and focus on interpretation.

Rules

- Use basic statistics only (no ML needed).
- Show formulas/steps, not just results.
- Where hypothesis tests are asked, clearly state **H0 and H1, method, assumptions, test statistic, p-value, and decision.**

Grading Rubric (100 points)

A. Funnel analytics & growth math — **20**
B. Webinar funnel & correlation — **15**
C. A/B test (setup + math + interpretation) — **15**
D. Satisfaction & NPS (calc + CI + actions) — **15**
E. B2B pipeline math & capacity insight — **10**
F. Retention analysis & regression — **10**
Short answers (18–20) — **10**
Presentation (clarity, charts, formatting) — **5**

Dataset A — Robo Ai Lead-to-Enrollment Funnel (Jan–Jun 2025)

Monthly figures are aggregated across campaigns.

Month	Website Visits	Leads (Form Fills)	Qualified Leads	Trial/Orientation Attendees	Enrollments	Revenue (₹)
Jan	18,000	1,260	720	520	200	28,00,000
Feb	20,500	1,435	820	610	250	36,25,000
Mar	23,200	1,740	960	700	280	40,60,000
Apr	24,100	1,810	1,020	760	300	43,50,000
May	25,600	1,940	1,110	810	320	46,40,000
Jun	31,000	2,480	1,380	1,020	400	58,00,000

Assume **list price ₹14,500 per enrollment; occasional scholarships/discounts explain revenue variance.**

Tasks (A)

1. **Compute stage-wise conversion rates each month:**

- Visits → Leads
- Leads → Qualified
- Qualified → Orientation
- Orientation → Enrollments
- Visits → Enrollments

Present **a table + a funnel chart for Jan and Jun.**

2. **Compute MoM growth** for **Enrollments and Revenue** and the **Jan→Jun CAGR for Enrollments.**

Identify the **single biggest month-over-month jump** and provide **one likely operational reason.**

3. Using a simple forecast (**either 3-month moving average or Holt linear trend—state which**), forecast **July 2025 Enrollments.**
4. Given estimated **variable delivery cost ₹3,200 per enrolled learner**, estimate:
- **Gross contribution per month**
- **Contribution margin % (Revenue basis)**

Briefly comment on **sustainability.**

Dataset B — Robopack Webinar Funnel (4 Events)

Event Date (2025)	Registrations	Show-Ups	Avg Watch Time (mins)	CTA Clicks	Conversions (Paid)
Feb 22	2,400	1,140	26	310	86
Mar 29	2,700	1,260	28	360	101
May 10	3,100	1,420	31	420	130
Jun 21	3,600	1,620	33	510	160

Tasks (B)

5. Compute stage conversions for each event:
- Reg → Show
- Show → Click
- Click → Paid
- Reg → Paid
6. Test whether **increased watch time is associated with higher Reg → Paid conversion.**

Use **Pearson correlation (state r and interpret).**

7. If each webinar costs **₹90,000 all-in (ads, tools, speaker)**, estimate:
- **Cost per Paid**
- **Simple ROAS**

Using **list price ₹1,499.**

Dataset C — Pricing A/B Test (Robo Ai Landing Page)

Two cohorts ran in parallel (**same traffic sources and dates**).

Variant	Visits	Leads	Enrollments	Price (₹)	Scholarships Given
A (List ₹14,500)	12,000	840	210	14,500	0
B (Anchor 16,500 → Show Offer 14,500)	12,300	930	246	14,500	0

Tasks (C)

8. Test if **Variant B improved the visit → enrollment rate vs A**.

Use a **two-proportion z-test at α = 0.05.**

Report:

- **z**
- **p**
- **decision**

9. Compute **Lead → Enrollment conversion for both variants.**

Offer **one UX hypothesis** for the observed difference.

Dataset D — Student Satisfaction & NPS (Robo Ai, June 2025 Cohort)

Ratings (1–5)

5, 4, 5, 3, 4, 4, 5, 2, 3, 4,
5, 5, 4, 3, 5, 4, 4, 5, 3, 5,
2, 4, 5, 5, 4, 3, 4, 5, 4, 5

NPS Question (0–10): sample of 40 responses

10, 9, 8, 7, 10, 6, 9, 8, 9, 10,
5, 6, 7, 8, 9, 9, 4, 10, 8, 9,
10, 3, 9, 8, 7, 10, 9, 6, 8, 9,
10, 8, 9, 7, 6, 9, 8, 10, 9, 8

Tasks (D)

10. Build a **frequency distribution of ratings**; compute:
- Mean
- Median
- Mode
- SD
11. Compute **NPS (Promoters 9–10, Passives 7–8, Detractors 0–6).**

Provide:

- **NPS score**
- **90% CI using a simple normal approximation**
12. Translate findings into **two product/ops actions for the next cohort.**

Dataset E — B2B College Outreach (May–Jul 2025)

Month	Emails Sent	Replies	Meetings	MoUs Signed	Seats Committed	Est. Realization Rate
May	900	72	24	3	240	70%
Jun	1,100	77	28	4	360	65%
Jul	1,300	91	30	5	420	68%

Tasks (E)

13. Compute **stage-wise B2B conversions** and identify the **biggest drop-off stage.**
14. Estimate **expected realized seats per month** using the **realization rate** and discuss how this affects **inventory planning for mentors/labs.**

Dataset F — LMS Engagement & Retention (Robo Ai, Jun 2025 Intake)

Week	Active Users	Avg Sessions/User	Avg Minutes/Session	Assignments Submitted
W1	400	3.2	24	380
W2	385	3.0	23	360
W3	360	2.8	22	340
W4	335	2.6	21	320

Tasks (F)

15. Compute **W1 → W4 retention %** and a **simple churn rate.**
16. Test whether **Avg Minutes/Session has a linear trend across weeks** (fit a simple regression **Minutes ~ Week**; report **slope and R^2**).
17. Suggest **one intervention to improve retention backed by the data.**

Short Answer (Concepts in the My Equation Context)

1. When **enrollment revenue grows faster than enrollments**, what statistical/operational explanations could fit our context?
   Give **two with one validation metric each.**
2. In **highly skewed lead quality distributions**, which central tendency (**mean / median / mode**) would you report to leadership and why?
3. Outline a **minimal pre-post experiment** to test if adding a **30-minute mentor AMA during orientation** improves **Orientation → Enrollment conversion.**

Output Expectations & Formatting

Include at least **4 charts**:

1. Funnel (**Jan & Jun**)
2. MoM Growth (**Enrollments**)
3. Webinar conversions vs **Watch Time**
4. **NPS breakdown**

Formatting rules:

- All **percentages to one decimal place**
- Currency in **₹ lakhs where sensible**
- Clearly **label assumptions**
